

**3: Oblique shocks and waves**

**3: Solving for conical  $\beta$**

AME 323: Gas Dynamics

# Solving Taylor-Maccoll numerically

First, it helps to define a dimensionless velocity

$$V' = \frac{V}{V_{max}}$$

Taylor-Maccoll is now

$$\frac{\gamma - 1}{2} \left[ 1 - V_r' - \left( \frac{dV_r'}{d\theta} \right)^2 \right] \left( 2V_r' + \frac{dV_r'}{d\theta} \cot \theta + \frac{d^2 V_r'}{d\theta^2} \right) - \frac{dV_r'}{d\theta} \left[ V_r' \frac{dV_r'}{d\theta} + \frac{dV_r'}{d\theta} \left( \frac{d^2 V_r'}{d\theta^2} \right) \right] = 0$$

You can show that

$$V' = \left[ \frac{2}{(\gamma - 1)M^2} + 1 \right]^{-\frac{1}{2}}$$

We need a few tricks now to solve this...

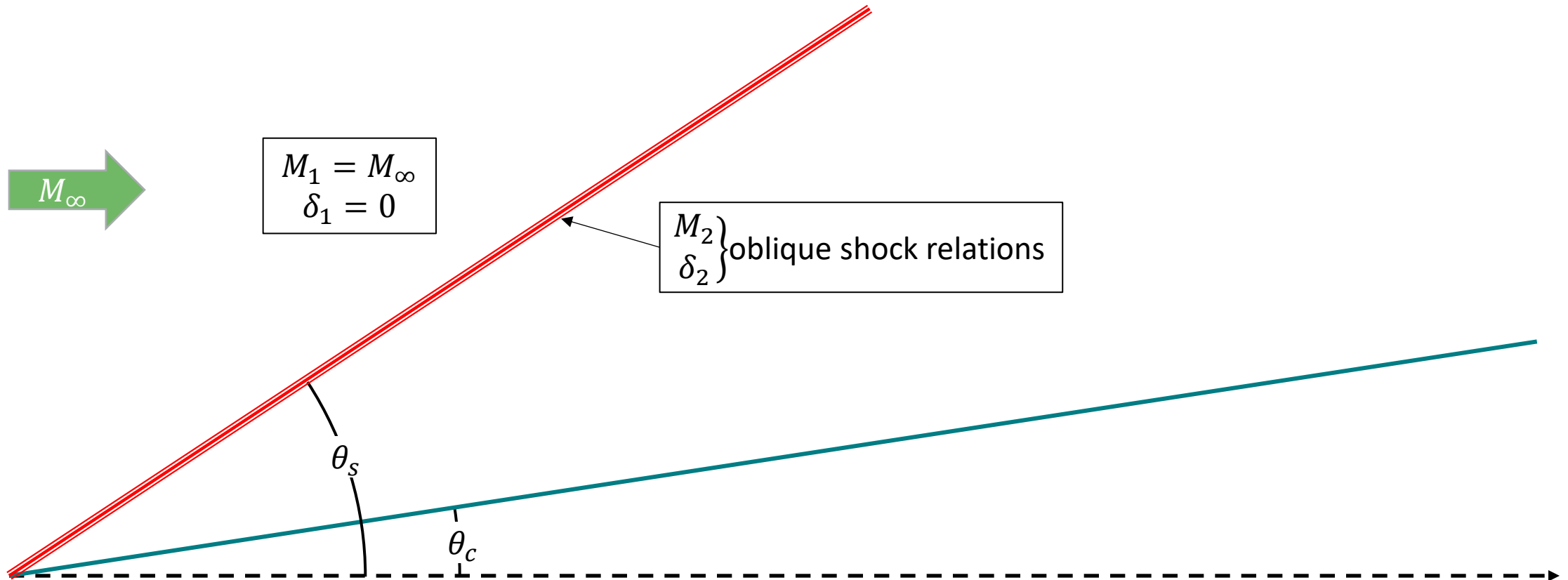
# Solving Taylor-Maccoll numerically

Typically, we employ the **shooting method**, i.e. guess an IC (in this case the shock angle,  $\beta$ ) and integrate toward a known BC (in this case the wall,  $\delta_c$ ).

1. Assume a shock angle  $\theta_s = \beta$  and a given  $M_\infty$  and use oblique shock relations to find  $\delta$  and  $M_2$  behind the assumed shock.
2. Determine  $V'$  at  $\theta_s$  from  $M_2$  and break it down into  $r$  and  $\theta$  components using  $\delta$ .
3. The value of  $V'_r(\theta_s)$  from step 2 is a boundary value for the flow (the other is  $V'_\theta(\theta_c) = 0$  to enforce “no penetration”).
4. Use any standard numerical technique (e.g. Runge-Kutta) to solve the governing equation from  $\theta_s$  until  $\theta_0$  where  $V'_\theta = 0$ , representing the wall that produced our assumed  $\beta$ .
5. Compare  $\theta_0$  found in step 4 to  $\theta_c$  specified in the problem. Adjust  $\beta$  accordingly and redo steps 1–4 until  $\theta_0 = \theta_c$ .

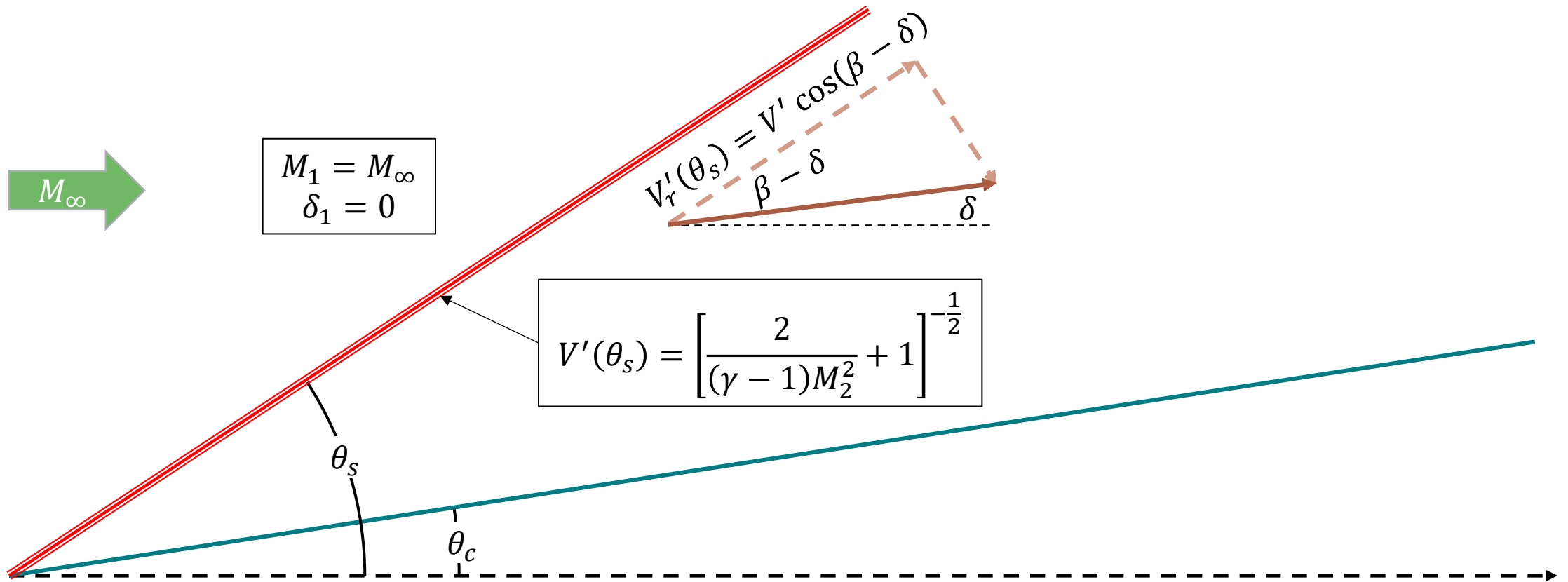
# Step 1

Given  $M_\infty$  and  $\theta_c$ , assume a shock angle  $\theta_s$ . Use standard oblique shock relations to find  $\delta_2$  and  $M_2$  immediately behind the assumed shock. A good guess for  $\theta_s$  would be between  $\mu$  and  $\beta$  from 2D wedge.



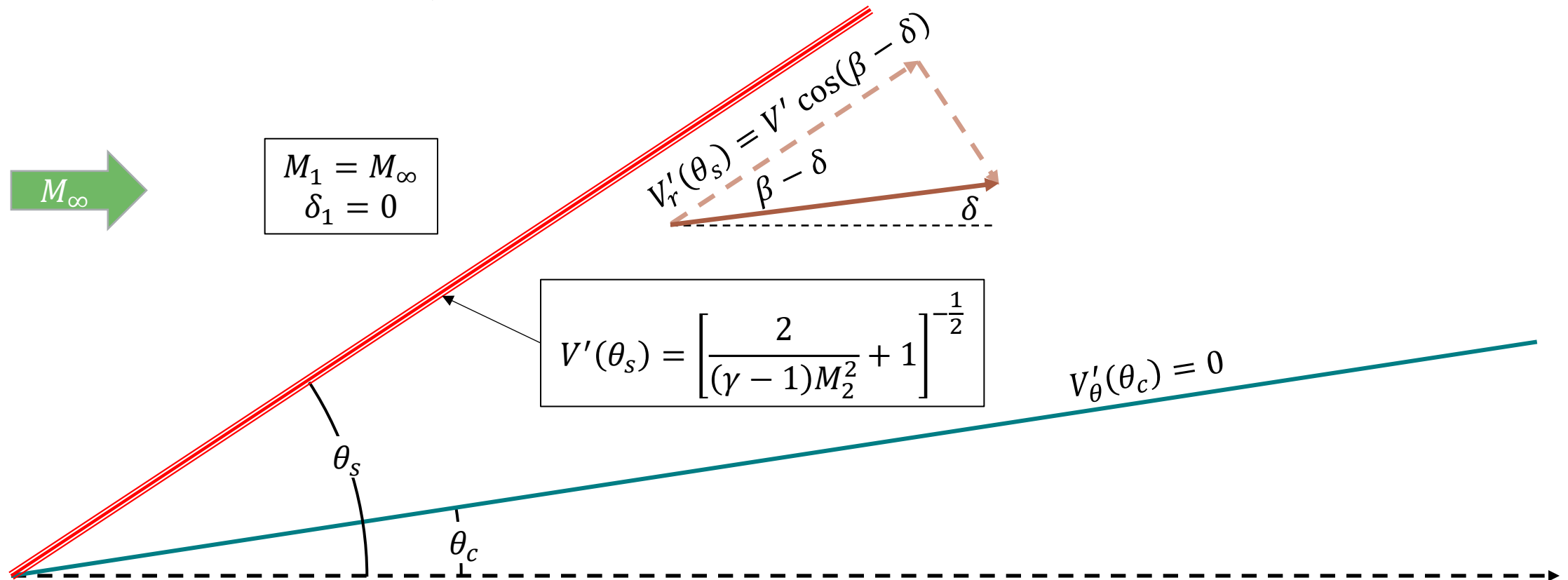
# Step 2

Determine  $V'$  at  $\theta_s$  from  $M_2$  and break it down into  $r$  and  $\theta$  components using  $\delta$ .



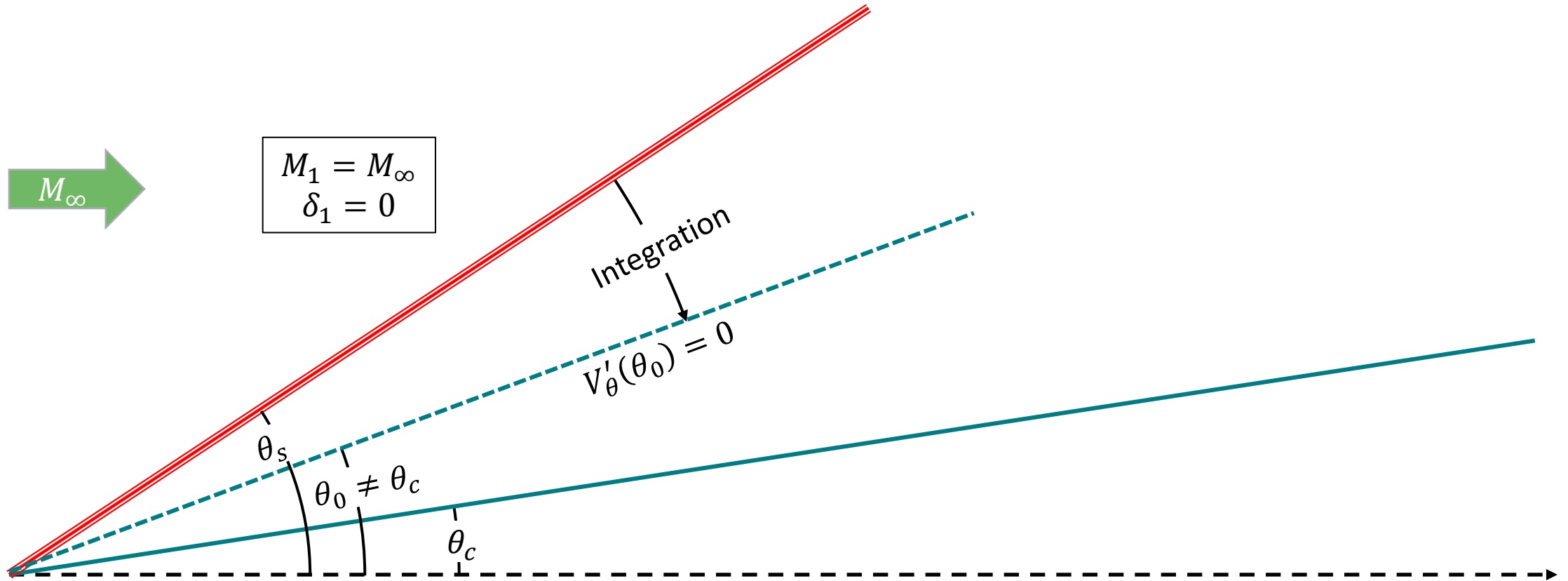
# Step 3

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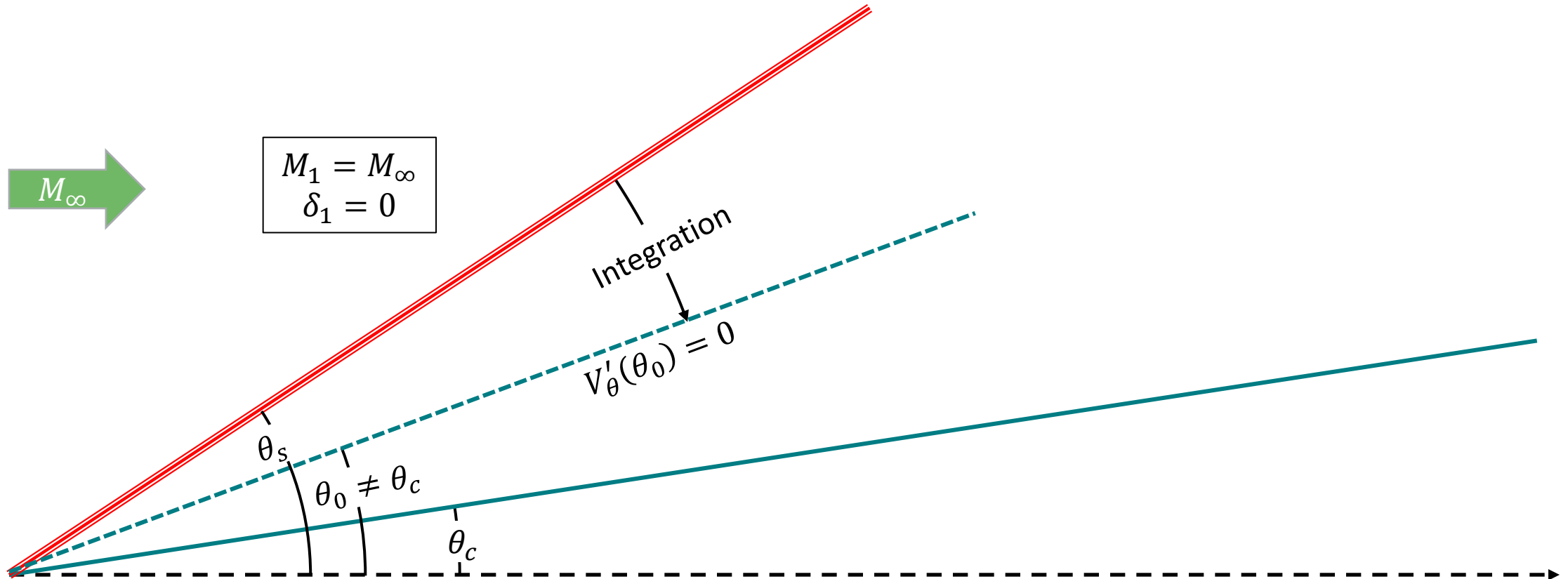
# Step 4

Use any standard numerical technique (e.g., Runge-Kutta) to solve the governing equation from  $\theta_s$  until  $\theta_0$  where  $V'_\theta = 0$ , representing the wall that produced our assumed  $\beta$ .



# Step 5

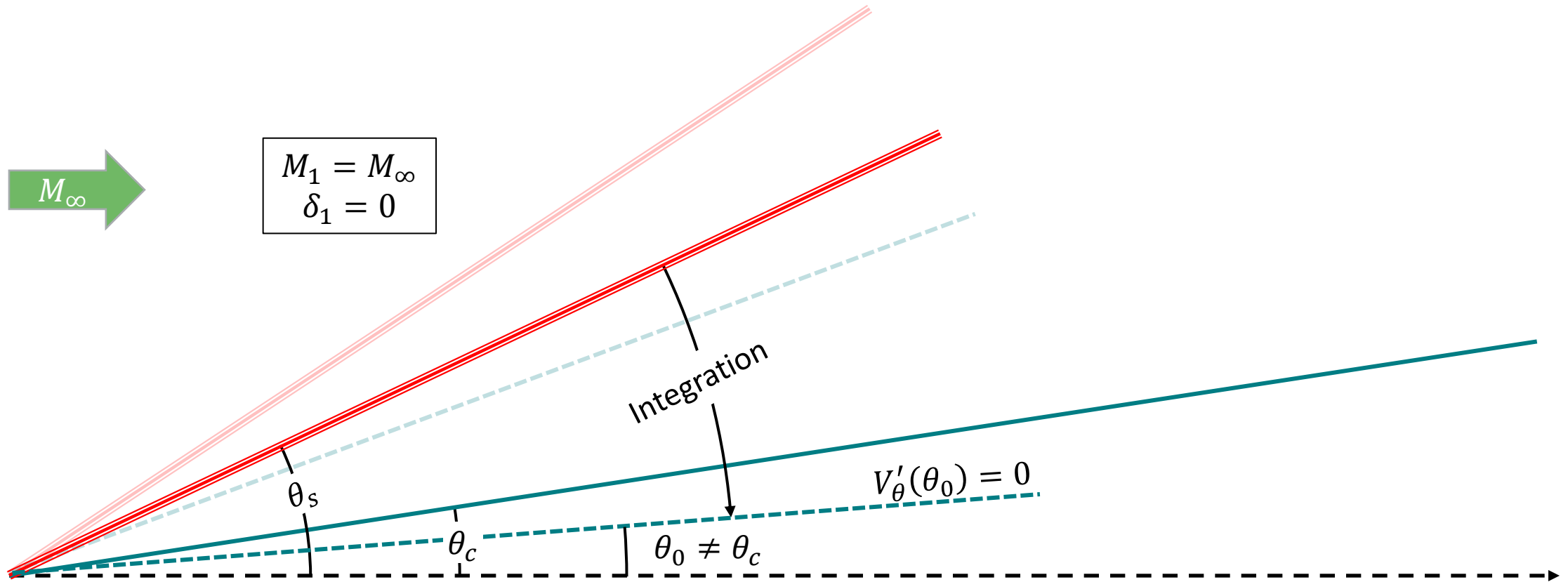
1. Compare  $\theta_0$  found in step 4 to  $\theta_c$  specified in the problem. Adjust  $\theta_s$  accordingly and redo steps 1—4 until  $\theta_0 = \theta_c$ . (e.g., bisection method)





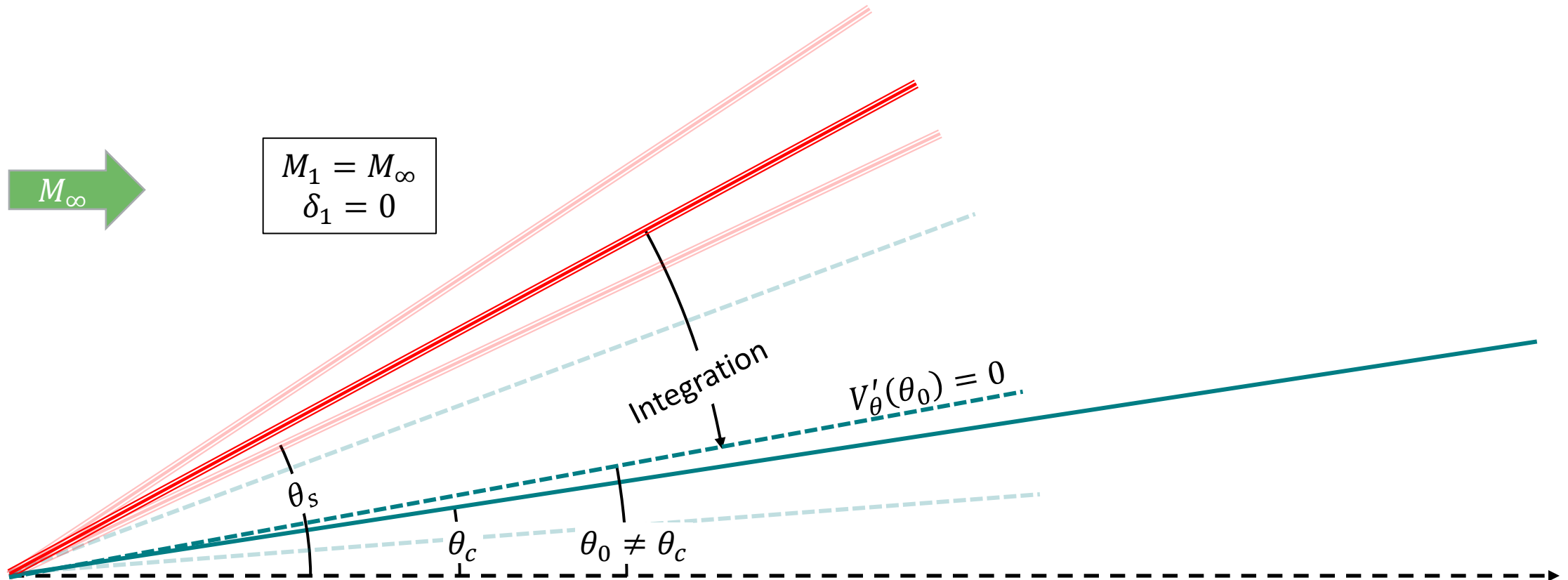
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